

**Emergent electrostatics of quantum spin ice:
measuring the permittivity, the speed of light and the fine
structure constant from electric-flux sectors**

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Abstract

[SUPERSEDED BY ITS OWN TEST — see README.md and Sec. VI. The quadratic law proposed below FAILS at every coupling, cluster and virtual depth measured: 29 of 29 fits, best residual 21.5%. The text below describes the calculation as planned; the measurement returned a null.]

A deconfined compact $U(1)$ gauge theory on a three-torus carries 't Hooft electric-flux sectors whose ground energies obey an emergent electrostatics, $\Delta E(\mathcal{T}) = |\mathcal{T}|^2/2\varepsilon_{\text{eff}}L$, with a single renormalized permittivity. Quantum spin ice realizes exactly this structure — the flux label is the total polarization of the ice configuration — but a periodic cluster cannot ordinarily be used to measure it, because the finite lattice supports virtual processes that tunnel between flux sectors and destroy the label. Removing those processes by transport masking makes the sectors exact and turns a 32-site cluster into an electrostatics experiment. We resolve the ground energy of all 85 flux sectors of the 32-site FCC cluster at thirteen couplings. The sector energies organize into seven shells whose members are degenerate to 1.5×10^{-15} ; the lowest excited shell lies $1.374 \times 10^{-3} J_{zz}$ above the zero-flux ground state at $J_{\pm}/J_{zz} = -0.05$. From the small-flux shells we extract ε_{eff} , and combining it with the independently measured nonperturbative ring amplitude K_6 gives the emergent light speed and hence the fine structure constant α — for the π -flux background, which is inaccessible to sign-free quantum Monte Carlo. The extraction is overdetermined: the same c must reproduce the directly computed excitation energies at the X and L points, giving three independent routes to one number. We further find that the flux-sector splitting *shrinks* with cluster size at weak coupling, as the Coulomb phase requires, but *grows* by factors of 2.8–3.5 at $J_{\pm}/J_{zz} = -0.22$ and -0.30 , which locates the breakdown of emergent electrostatics on the XXZ line.

I. INTRODUCTION

The emergent electrodynamics of quantum spin ice is usually characterized by two numbers: the speed of light c of the emergent photon and the fine structure constant α that measures how strongly that photon couples to the emergent charges [1–3]. The second is the more interesting and the less well determined. Unlike its counterpart in nature, α in quantum

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spin ice is predicted to be *large* [4], of order unity or above over much of the phase diagram, which is why the emergent photon is strongly renormalized and why photon–photon and photon–spinon effects are expected to be observable at all. The available determinations are perturbative or variational.

The obstruction to a nonperturbative determination is not one of raw computing power. Sign-free quantum Monte Carlo is restricted to $J_{\pm} > 0$, the 0-flux side, whereas every candidate material sits at $J_{\pm} < 0$, in the π -flux background [5–7]. Exact diagonalization works at both signs, but the standard route — fit a photon dispersion to a finite-cluster spectrum — fails because the accessible momenta are a handful of zone-boundary points and nothing is converged in system size.

This paper takes a different observable, one that is a *ground-state* property and therefore far better behaved on small clusters: the energy of the electric-flux sectors. In a deconfined phase these energies are not accidental numbers; they are fixed by the emergent Maxwell action, and their pattern measures the permittivity directly. The construction requires one ingredient that has only recently become available: the flux sectors must be *exact*, which on a finite torus they are not.

II. ELECTRIC-FLUX SECTORS OF QUANTUM SPIN ICE

A. The label

We study the nearest-neighbour XXZ model on the pyrochlore lattice,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + \text{h.c.}), \quad (1)$$

with $J_{zz} > 0$. At $J_{\pm} = 0$ the ground manifold is the two-in–two-out ice manifold, on which the emergent electric field is $E_i = S_i^z$ and the ice rule is Gauss’s law. An ice configuration a carries the polarization

$$\mathbf{p}_a = \sum_i S_i^z(a) \mathbf{r}_i, \quad \mathcal{T}_a = 2\mathbf{p}_a, \quad (2)$$

with $\mathbf{r}_i$ the lifted site position. $\mathcal{T}$ is the total electric flux through the three cross-sections of the torus, i.e. the ’t Hooft electric-flux sector label [8], and it is the finite-volume remnant of the one-form symmetry of the deconfined phase [9]. On the 32-site FCC cluster (FCC-32) the 2970 ice configurations fall into 85 sectors, with dimensions from 1 to 396.

B. Why a naive cluster has no flux sectors

A contractible rearrangement of spins cannot change $\mathcal{T}$; a rearrangement that closes only through a periodic image changes it by a box vector. The infinite lattice has only the first kind, so $\mathcal{T}$ is exactly conserved in the thermodynamic limit and the flux sectors are superselected. A finite periodic cluster has both. Worse, on the clusters that are actually diagonalizable the flux-changing process is *shorter* than the physical one: on FCC-32 a four-spin winding path appears at order $J_{\pm}^2/J_{zz}$, one order before the hexagon ring exchange at $12|J_{\pm}|^3/J_{zz}^2$ [10]. The naive periodic ground state is therefore a coherent superposition of all flux sectors and the sector energies do not exist as observables.

This is why the measurement proposed here has not been made before, and it is also why it is now possible. We use the transport-masked Feshbach map of Ref. [10]: virtual spinon excursions are summed exactly through $F(z) = PHP + PHQ(z - QHQ)^{-1}QHP$, and every matrix element connecting different $\mathcal{T}$ is deleted. The resulting theory is block diagonal in $\mathcal{T}$ by construction, each block is solved as a bordered Hermitian problem whose Schur complement is the masked block, and the ground energy of each of the 85 blocks is a well-defined number.

C. What the emergent theory predicts

Coarse-graining Eq. (1) on the ice manifold gives the emergent Maxwell energy

$$\mathcal{E} = \int d^3r \left[\frac{1}{2\varepsilon} |\mathbf{D}|^2 + \frac{1}{2\mu} |\mathbf{B}|^2 \right], \quad c = \frac{1}{\sqrt{\varepsilon\mu}}. \quad (3)$$

A uniform flux $\mathcal{T}$ through a box of linear size L corresponds to $|\mathbf{D}| = |\mathcal{T}|/L^2$ and volume L^3 , so

$$\boxed{\Delta E(\mathcal{T}) \equiv E_0(\mathcal{T}) - E_0(\mathbf{0}) = \frac{|\mathcal{T}|^2}{2\varepsilon_{\text{eff}}L}} \quad (4)$$

in the deconfined phase. Three features of Eq. (4) make it a good test rather than a fit. It is *isotropic*: ΔE may depend on $\mathcal{T}$ only through $|\mathcal{T}|$, so sectors in different point-group stars that happen to share $|\mathcal{T}|$ must be degenerate — a nontrivial prediction the lattice symmetry does not by itself require. It is *quadratic*, with a single coefficient for all shells. And its size dependence is $1/L$: the splitting must *shrink* as the cluster grows, in contrast to a confining phase where it would grow linearly.

TABLE I. FCC-32 electric-flux sector ground energies, $J_{\pm}/J_{zz} = -0.05$, virtual depth $d = 1$. “mult” is the number of sectors in the shell, “dim” the ice-manifold dimension of each. Energies in units of J_{zz} ; ΔE is measured from the zero-flux sector. All 85 sectors and 2970 ice states are accounted for ($\sum \text{mult} = 85$, $\sum \text{mult} \times \text{dim} = 2970$).

mult	dim	ΔE ($10^{-3} J_{zz}$)	ratio to first
1	396	0.000000	—
12	140	1.373749	1.000
6	72	1.896027	1.380
24	12	6.722887	4.894
12	6	6.729408	4.899
24	4	6.896969	5.020
6	1	7.090524	5.161

One physical remark is essential to interpreting what follows. On the ice manifold the Ising energy is the same for every configuration and every link carries $|E| = 1/2$, so $\sum_{\text{links}} E^2$ is a constant and contributes nothing to Eq. (4). **The entire emergent electric stiffness is generated dynamically by the ring exchange:** a polarized sector has fewer flippable hexagons and therefore loses resonance energy. Consequently the observation that ΔE increases as the sector dimension falls is not an alternative, “entropic” explanation competing with electrostatics — it is the microscopic mechanism *of* the emergent electrostatics. What distinguishes the two readings is quantitative and is exactly the content of Eq. (4): whether ΔE is a function of $|\mathcal{T}|^2$ alone, with one coefficient.

III. SECTOR SPECTROSCOPY OF FCC-32

A. The shell structure

Table I lists the ground energy of every one of the 85 sectors at $J_{\pm}/J_{zz} = -0.05$, virtual depth $d = 1$ (a common 140 442-state microscopic space). The 85 numbers collapse onto *seven* distinct values. Within each shell the sector ground energies agree to 1.5×10^{-15} , i.e. to solver precision: the degeneracy is exact, not approximate.

The multiplicities 1, 12, 6, 24, 12, 24, 6 are exactly the star sizes of the cubic point group, so each shell is a single orbit of $\mathcal{T}$ under the space group. The unique 1-dimensional shell of multiplicity 6 is the set of fully polarized “frozen” sectors, which contain one configuration each and no ring exchange at all; the unique 396-dimensional sector is $\mathcal{T} = \mathbf{0}$ and is the global ground state.

B. The two tests Eq. (4) must pass

a. Isotropy. Two of the seven shells have multiplicity 24 and two have multiplicity 12. If any two distinct point-group stars share the same $|\mathcal{T}|$, Eq. (4) requires them to be degenerate. Table I shows the two multiplicity-12 shells at 1.374 and 6.729, and the two multiplicity-24 shells at 6.723 and 6.897 — the latter pair differing by only 2.6% while the former differ by a factor of five. **[PENDING: the assignment of $|\mathcal{T}|$ to each shell requires joining the sector indices to the polarization vectors, which is pure combinatorics on the 2970 ice states (`protocol.polarization_sector_labels`); until it is done we cannot say whether the near-degenerate multiplicity-24 and multiplicity-12 pairs share $|\mathcal{T}|$, which is the isotropy test. Script: `analysis/sector_labels.py`.]**

b. Quadraticity. The observed ratios 1.000, 1.380, 4.894, 4.899, 5.020, 5.161 do *not* continue to grow across the full set: after the third shell they saturate, changing by only 5% across the last four. A pure $|\mathcal{T}|^2$ law with shells at increasing $|\mathcal{T}|$ cannot do this. We therefore expect Eq. (4) to hold only in a weak-field window — plausibly the first two or three shells — with the saturation marking the field strength at which the compact lattice theory departs from its continuum limit. This is a physically sensible outcome on a cluster only two conventional cells across, where the largest sectors are fully polarized, but it must be stated as a limitation of the extraction and not hidden. **[PENDING: the fit of Eq. (4), and hence ε_{eff} , awaits the $|\mathcal{T}|$ join.]**

C. Coupling and depth dependence

Sector ground energies are available at thirteen couplings from $J_{\pm}/J_{zz} = -0.05$ to -0.32 at depth $d = 1$, and at four couplings ($-0.05, -0.22, -0.26, -0.30$) at depth $d = 2$, which enlarges the virtual space from 1.4×10^5 to 2.5×10^6 states and the cost from 51 s to

5.8×10^3 s per coupling. The depth sensitivity is not negligible: at $J_{\pm}/J_{zz} = -0.05$ the first shell gap moves from 1.374×10^{-3} at $d = 1$ to 1.837×10^{-3} at $d = 2$, an increase of 34%. Since $\varepsilon_{\text{eff}} \propto 1/\Delta E$, the permittivity carries at least this systematic, and any quoted α must be accompanied by the depth ladder. **[PENDING: the d -dependence of ε_{eff} at all four matched couplings.]**

IV. FROM ε_{eff} TO c AND α

The magnetic constant of the emergent theory is set by the ring-exchange amplitude. In the effective gauge Hamiltonian $H_{\text{eff}} = -K \sum_{\text{hex}} \cos B + \frac{1}{2} \varepsilon^{-1} \sum |\mathbf{D}|^2$ one has $\mu^{-1} \propto K$, so with ε_{eff} from Sec. III and K measured independently,

$$c = \kappa_c a \sqrt{\frac{K}{\varepsilon_{\text{eff}}^{-1}}} / \hbar, \quad \alpha = \kappa_{\alpha} \sqrt{\frac{1}{\varepsilon_{\text{eff}} K}}, \quad (5)$$

where a is the lattice constant and $\kappa_c, \kappa_{\alpha}$ are pure lattice geometry factors, fixed once and for all by matching Eq. (5) to the analytically known weak-coupling limit [1]. Crucially, K here is not the perturbative $g_6 = 12|J_{\pm}|^3/J_{zz}^2$ but the measured nonperturbative amplitude of the masked map, which the same construction supplies: the hexagon element of the folded operator, whose ratio to g_6 falls from 0.81 at $J_{\pm}/J_{zz} = -0.02$ to 0.07 at -0.35 . Using the bare g_6 would misestimate α by an order of magnitude at strong coupling.

A. An overdetermined check

Equation (5) can be tested rather than merely evaluated, because c is also directly observable in the same calculation as the slope of the excitation energies at the cluster momenta. The masked spectrum gives, at depth $d = 1$,

$J_{\pm}/J_{zz}$	$\omega(\Gamma)$	$\omega(X)$	$\omega(L)$
+0.046	0.002922	0.003612	0.002217
-0.200	0.018874	0.025274	0.015678

so we have three routes to one number: ε_{eff} from sector energies, K from loop tomography, and $\omega(\mathbf{q})$ from the spectrum. If Eq. (5) reproduces the measured $\omega(L)$ and $\omega(X)$ with one κ_c , the emergent Maxwell description is validated nonperturbatively in both flux backgrounds. If it does not, the discrepancy measures how far FCC-32 is from the continuum limit, which is

itself worth quantifying: the observed ratio $\omega(X)/\omega(L)$ is 2.04 and 2.42 at the two couplings against the linear-dispersion value $|X|/|L| = 2/\sqrt{3} = 1.155$, so we expect the dispersion route to be the weakest of the three. **[PENDING: all of Sec. IV is a plan; no value of c or α has been computed.]**

V. WHERE EMERGENT ELECTROSTATICS ENDS

Equation (4) predicts that the flux-sector splitting falls as $1/L$. Comparing the 16-site cubic cluster with FCC-32 at matched coupling, the splitting is multiplied by 0.65 at $J_{\pm}/J_{zz} = -0.05$ — close to the $(N_{32}/N_{16})^{-1/3} = 0.79$ expected for a $1/L$ law and clearly a shrinkage — but by 3.5 and 2.8 at $J_{\pm}/J_{zz} = -0.22$ and -0.30 . A splitting that *grows* with system size is not the behaviour of a Coulomb phase.

Read at face value this locates the breakdown of emergent electrostatics on the XXZ line between $J_{\pm}/J_{zz} = -0.05$ and -0.22 , which is compatible with, and sharper than, the range $[-0.25, -0.5]$ that other diagnostics of the same campaign leave open for the end of the π -flux phase. It is also consistent with two independent observations: the masked hexagon flux is $-1/4$ to sixteen digits through $J_{\pm}/J_{zz} = -0.24$ and drops to exactly 0 from -0.26 , and the ice-descended band ceases to be isolated from the spinon continuum around the same place.

Two caveats keep this from being a phase-boundary claim. Only two clusters exist, so “scaling” means a single ratio. And the two clusters are not equivalent hosts: the 16-site cluster has six degenerate ground sectors against FCC-32’s one, and it resolves no L point at all. **[PENDING: a third host is required to make this a scaling statement; the geometry builder supports rectangular FCC boxes, so an FCC $2 \times 2 \times 3$ (48 sites) cluster is the natural next point, subject to the sector solver’s cost.]**

VI. FALSIFICATION: THE MEASUREMENT, AND WHAT IT RETURNED

The tests deferred in Sec. III have now been run (`analysis/cubic16_sector_check.py`, job 19155241). The transport labels were joined to the sector indices on both clusters, and the cubic-16 sector ground energies were recomputed at virtual depths $\ell = 1, 2, 3, 4$, where $\ell = 4$ is the *complete* space and therefore truncation-free. The outcome is a null, and it is

not marginal.

a. The transport labels. FCC-32 has seven distinct $|\mathcal{T}|^2$ values, 0, 2, 4, 6, 8, 10, 16, with multiplicities 1, 12, 6, 24, 12, 24, 6 and ice dimensions 396, 140, 72, 12, 6, 4, 1 — confirming the shell assignment of Table I and identifying the 396-dimensional ground sector as $\mathcal{T} = 0$. Cubic-16 has four, 0, 1, 2, 4, with multiplicities 1, 6, 12, 6 and dimensions 12, 8, 2, 1.

b. Quadraticity fails everywhere. Fitting $\Delta E = |\mathcal{T}|^2/2\varepsilon_{\text{eff}}L$ to the three smallest nonzero fluxes gives a maximum relative residual of 0.215 at best and 0.441 at worst, over 29 fits spanning both clusters, four virtual depths and the couplings $J_{\pm}/J_{zz} = -0.05, -0.20, -0.30$ (cubic-16, exact) and -0.05 through -0.32 (FCC-32). *Not one fit passes.* On FCC-32 at $J_{\pm}/J_{zz} = -0.05$, depth 1, the law would require $\Delta E(|\mathcal{T}|^2 = 4) = 2\Delta E(|\mathcal{T}|^2 = 2)$; the measured ratio is 1.38. Between $|\mathcal{T}|^2 = 4$ and 6 the energy then jumps by a factor of 3.5 and is flat thereafter.

c. What the data show instead. The sector energies separate into two families: large sectors (dim = 396, 140, 72) that still support substantial ring resonance, and small ones (dim = 12, 6, 4, 1) that are nearly frozen, with a step between them and near-degeneracy within the second family. This is the signature of a lattice ring-exchange model whose polarized sectors have simply run out of flippable plaquettes — the discrete, strong-field regime — not of a Maxwell field in its continuum limit. The mechanism identified in Sec. II is correct; the continuum law built on it is not reached at these sizes.

d. The isotropy test is vacuous here. On both clusters every $|\mathcal{T}|^2$ value corresponds to exactly one cubic point-group star: no two distinct stars share a modulus. The exact degeneracies observed within each shell ($\leq 6 \times 10^{-14}$ over the whole run) are therefore guaranteed by the space group and carry no information about emergent isotropy. This removes what was to have been the sharpest structural test.

e. Zero flux is not always the ground. On cubic-16 the *exact* ($\ell = 4$) ground sector lies at $|\mathcal{T}|^2 = 1$, six-fold degenerate, for $J_{\pm}/J_{zz} = -0.05$ and -0.20 , and only moves to $\mathcal{T} = 0$ by -0.30 . This finally explains the six-fold ground multiplet recorded throughout the campaign: it is a flux-polarized ground state, not a resonance degeneracy. FCC-32, by contrast, has its ground in the unique $\mathcal{T} = 0$ sector at every coupling and depth measured. The two clusters disagree qualitatively about where the electrostatic minimum sits.

f. Depth truncation is severe. On cubic-16 at $J_{\pm}/J_{zz} = -0.30$ the identity of the ground sector itself is a truncation artifact: $\ell = 1$ places it at $|\mathcal{T}|^2 = 1$, whereas $\ell \geq 2$ and

the exact answer place it at $\mathcal{T} = 0$. On FCC-32 the $\ell = 1 \rightarrow 2$ change at $J_{\pm}/J_{zz} = -0.05$ is +34%, +31% and +21% for the first three shells, and the ordering of the $|\mathcal{T}|^2 = 6$ and 8 shells *inverts* between the two depths. Since all thirteen FCC-32 couplings are at $\ell = 1$, no shell spacing quoted here is converged.

g. Consequence. No weak-field window exists on either cluster in which ε_{eff} is defined, so the extraction of c and α described in Sec. IV cannot be carried out on these hosts. The size at which such a window opens is the open question this measurement replaces the original one with.

VII. LIMITATIONS

1. **Two cells per direction.** FCC-32 is $2 \times 2 \times 2$ conventional cells. The weak-field window in which Eq. (4) can hold is correspondingly narrow, and the saturation reported in Sec. III is direct evidence of that narrowness.
2. **Virtual-depth truncation.** ε_{eff} inherits at least the 34% $d = 1 \rightarrow 2$ drift of the first shell gap. Depth $d = 3$ on FCC-32 costs $\sim 2 \times 10^7$ states and is a ~ 250 GB job; $d = 4$ is out of reach.
3. **The geometry factors.** κ_c and κ_{α} in Eq. (5) must be fixed against the analytic weak-coupling limit before any absolute α is quoted; otherwise only ratios and trends are meaningful.
4. **The mask is the enabling assumption.** Everything here presumes that deleting transport-changing amplitudes is the correct finite-volume implementation of the one-form symmetry. That is argued and tested in Ref. [10]; this paper inherits the argument.

VIII. CONCLUSION

The energies of the 85 electric-flux sectors of a 32-site pyrochlore cluster are a direct, ground-state probe of the emergent electrostatics of quantum spin ice, and they become measurable once the torus artifacts that tunnel between flux sectors are removed. The sector energies collapse onto seven point-group shells that are exactly degenerate within

themselves, which is the first requirement of an emergent Maxwell description. Extracting the permittivity from the weak-field shells, and combining it with the nonperturbative ring amplitude, yields the emergent speed of light and fine structure constant on the π -flux side of the phase diagram, where sign-free quantum Monte Carlo cannot go; the same numbers are overdetermined by the directly computed excitation energies. Finally, the flux splitting shrinks with system size at weak coupling and grows at $J_{\pm}/J_{zz} \leq -0.22$, which is where the emergent electrostatics — and with it the Coulomb phase — ends on this lattice.

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